Experiment 10 : Closest Pair of Points – Brute Force

Aim

To write a Python program to find the closest pair of points in a 2D plane using the brute-force approach and analyze its time complexity.

Algorithm

1. Start.
2. Read a set of points in a 2D plane.
3. Initialize the minimum distance to infinity.
4. Compare every pair of points using two nested loops.
5. Calculate the Euclidean distance between each pair of points.
6. If the calculated distance is smaller than the minimum distance, update the minimum distance and closest pair.
7. Display the closest pair of points and their distance.
8. Stop.

Source Code

```
import math

def closest_pair(points):
    min_distance = float('inf')
    closest = None

    n = len(points)

    for i in range(n):
        for j in range(i + 1, n):
            x1, y1 = points[i]
            x2, y2 = points[j]

            distance = math.sqrt(
```

```
            (x2 - x1) ** 2 + (y2 - y1) ** 2
        )

        if distance < min_distance:
            min_distance = distance
            closest = (points[i], points[j])

    return closest, min_distance


points = [(2, 3), (12, 30), (40, 50), (5, 1), (3, 4)]

pair, distance = closest_pair(points)

print("Points:", points)
print("Closest Pair:", pair)
print("Minimum Distance:", round(distance, 2))
```

Output :

```
Output

Points: [(2, 3), (12, 30), (40, 50), (5, 1), (3, 4)]
Closest Pair: ((2, 3), (3, 4))
Minimum Distance: 1.41
```

Result :

Thus, the Python program was successfully executed, and the closest pair of points, (2, 3) and (3, 4), was identified using the brute-force approach with a minimum Euclidean distance of 1.41 units.